Manoj Kumar Ashok

(312) 284-9898 | mashok@depaul.edu | linkedin.com/in/MKASHOK | GitHub | Portfolio | Chicago, IL | Open to Relocate

Summary

M.S. Data Science candidate (DePaul, Mar 2026) with hands-on experience building machine learning models, statistical analysis pipelines, and data workflows in Python and SQL. Proven ability to deliver measurable results: 30% improvement in GAN model realism, 35% reduction in query latency, and classification models achieving 93.9% ROC-AUC. Experienced in the full ML lifecycle from data ingestion and feature engineering to model evaluation and reproducible reporting. Eager to apply rigorous analytical skills to real-world data science and ML problems.

Technical Skills

Languages: Python, SQL, R, C++
ML / Deep Learning: Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, CNNs, GANs, XGBoost, LightGBM
Data & Engineering: Pandas, NumPy, PySpark, ETL Pipelines, Airflow, Apache Spark, AWS (S3, Athena, EC2)
Statistics & Experimentation: Regression, Hypothesis Testing, A/B Testing, Causal Inference, Feature Engineering
Databases: MySQL, PostgreSQL, Hive, MongoDB
Tools: Git, Docker, Linux, CUDA, Tableau, Power BI, Matplotlib, Seaborn

Education

DePaul University — Chicago, IL
M.S. in Data Science — Computational Methods — *Jan 2024 – Mar 2026*

- **Relevant Coursework:** Advanced Machine Learning, Neural Networks & Deep Learning, NLP, Regression & Data Analysis, Mining Big Data, Database Processing for Large-Scale Analytics.

Bharathiar University — Coimbatore, India
Bachelor of Computer Applications — ML & AI — *Jul 2020 – Nov 2023*

- **Relevant Coursework:** Python Programming, Data Structures & Algorithms, AI & ML, Statistics, NLP, Computer Vision.

Professional Experience

Research Assistant — Machine Learning & Data Science — Jan 2025 – Mar 2026
DePaul University (Prof. David Ramsay) — *Chicago, IL*

- Researched and optimized Conditional GAN architectures (IcGAN, RoCGAN) for image generation; improved model realism by **30%** (FID/IS) and reduced training time by **40%** through GPU optimization and hyperparameter tuning.
- Built scalable Python-based ML pipelines for large imaging datasets, covering ingestion, preprocessing, validation, transformation, and reproducible experiment tracking.
- Applied causal inference techniques — including propensity score matching and difference-in-differences — to evaluate treatment effects in observational research settings.
- Cleaned and validated datasets across 5+ concurrent studies; developed automated QA workflows reducing manual review time by **25%** and ensuring analytical reproducibility.
- Evaluated models using precision, recall, F1, ROC-AUC, and confusion matrices; prepared results summaries and figures for investigator review and manuscript development.

Software Engineer — Data Engineering Team — Jan 2023 – Jul 2023
Zoho Corporation — *Chennai, India*

- Built and optimized ETL pipelines in Python and SQL processing millions of CRM and financial records daily across ingestion, transformation, and reporting layers; improved data throughput by **20%**.
- Reduced SQL query latency by **35%** through indexing, schema refactoring, and materialized view optimizations, enabling faster analytics for business stakeholders.
- Automated KPI dashboards and batch reporting workflows using SQL and Tableau, cutting reporting lag by **24%**.
- Implemented Airflow DAG orchestration for production pipelines, reducing batch latency by **15%** and improving system reliability.

Projects

GPU-Accelerated LLM Inference & RAG System | LLaMA.cpp, CUDA, Hugging Face, FastAPI, Docker *GitHub* — 2025

- Built a local LLM inference system with LLaMA.cpp + CUDA; improved inference latency by **30%** via kernel-level optimizations, quantization, and caching strategies.
- Implemented a retrieval-augmented generation (RAG) pipeline with document chunking and embedding-based retrieval; engineered prompt templates for classification and information extraction tasks.
- Evaluated model outputs using precision, recall, and F1-score; analyzed misclassifications and false positives to iteratively improve structured output quality.
- Containerized the full system with Docker for reproducible, scalable deployment; exposed modular REST API via FastAPI.

Heart Disease & Bankruptcy Risk Prediction | Python, Scikit-learn, XGBoost, LightGBM, SMOTE, SQL *GitHub* 2025

- Built ensemble classifiers (Gradient Boosting, MLP, Random Forest, Logistic Regression) on clinical data; achieved **88.6% accuracy** and **93.9% ROC-AUC** on heart disease prediction.
- Trained XGBoost/LightGBM ensemble on 6,800+ financial records for bankruptcy prediction; achieved **98.5% accuracy** and **97.6% F1-score**.
- Applied SMOTE oversampling and stratified sampling to address class imbalance, improving minority-class recall by up to **42%**.
- Used SQL to audit and validate input data quality; prepared results tables following clinical and financial research reporting standards.

NYC Taxi Tip Prediction & A/B Testing Engine | Python, PySpark, AWS Athena, SQL, NumPy 2024–2025

- Cleaned and feature-engineered 3M+ NYC taxi trip records using PySpark and AWS Athena; built regression models predicting tip percentage, improving baseline RMSE by **18%**.
- Designed a simulation framework generating synthetic experiment datasets to evaluate behavioral interventions; implemented hypothesis tests (t-test, chi-square, Bayesian inference) and computed lift, CTR, and retention metrics.
- Built SQL-backed evaluation pipelines to validate experiment data integrity and automate statistical reporting across study arms.

Skin Lesion Classification — HAM10000 | PyTorch, CNNs (VGG16, ResNet34), Scikit-learn *GitHub* 2025

- Implemented CNN architectures (VGG16, ResNet34) to classify 10,000+ dermoscopic images; improved accuracy from **65% to 84%** using regularization, augmentation, and hyperparameter tuning.
- Evaluated performance using ROC-AUC, F1-score, and confusion matrices; analyzed recall–precision tradeoffs for medical classification contexts.